

MEASURING TEMPERATURE

NS YEAR3 TERM 3

WE HAVE THE FOLLOWING

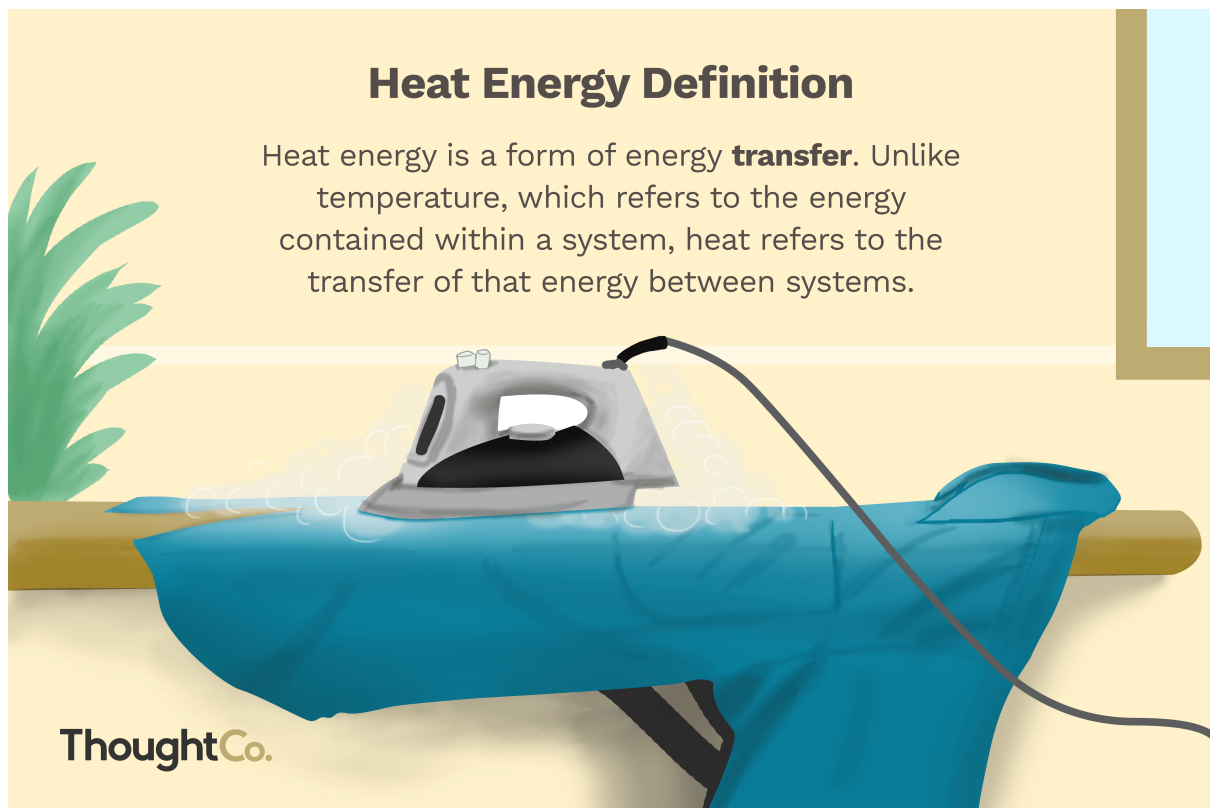
Types of Thermometer



THEN WE HAVE TO EXPLAIN ALL THE CONCEPTS UNDER ENERGY AND TEMPERATURE

HEAT, TEMPERATURE, CELCIOUS SCALE, BOILING POINT, MELTING PONT, CONDUCTOR, INSULATOR, THEMOMETER

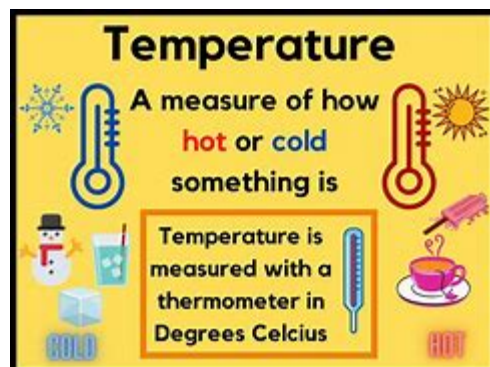
NOTE:THE LEARNER HAS TO UNDERSTAND DEFINATION AND MEANING OF THE ABOVE CONCEPTS ONLY THEN THEY CAN MATCH THE CONCEPT WITH THE DESCRIPTION



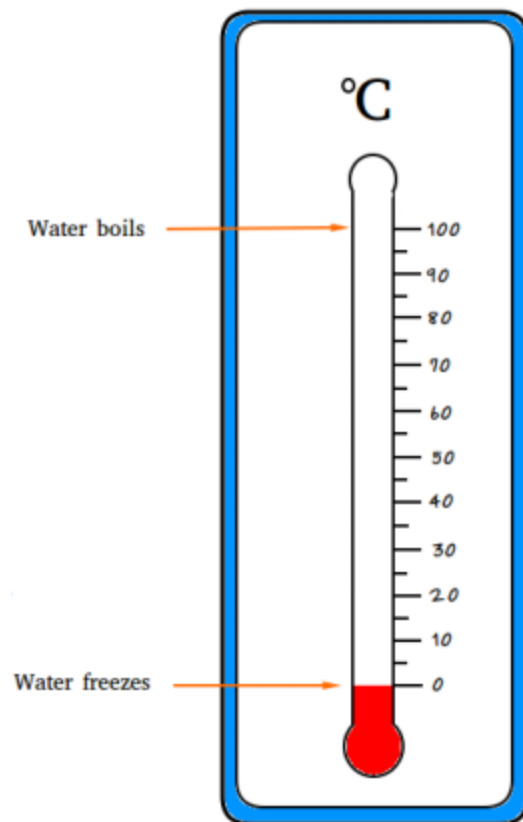
HEAT: Is the transfer of energy from one body to another as a result of a differences in temperature. When two bodies of differing temperature(warm/cold) come together energy is transmitted, and heat is transferred (--) from (hotter) to (colder)

3WAYS HOW ENERGY CAN BE TRANSFERRED

1.Conduction 2. Convection 3. Radiation



The Celsius scale is the measure of how (hot) or (cold) a substance feels and its measured in Celsius (c)



BOILING POINT

Boiling Point

The boiling point is the temperature at which a liquid changes into a vapor (boils).

Boiling Point of Water

at sea level
100 °C (212 °F)

at 6250 feet
93.4 °C (200.1 °F)



At the boiling point, the vapor pressure of the liquid is the same as the vapor pressure of air (or whatever is outside the liquid).

MELTING POINT

THE MELTING POINT



Melting point is the temperature at which a solid substance changes its state to become a liquid.

It is the temperature at which the thermal energy (heat) is sufficient to break the intermolecular forces that hold the particles of a solid together, allowing them to move freely and take on the properties of a liquid.

The **melting point** is a physical property of a substance and can be used to identify and characterize it.

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CONDUCTOR

Definition

Conductor

A **conductor** is an object or type of material that allows the flow of an **electrical current** in one or more directions.

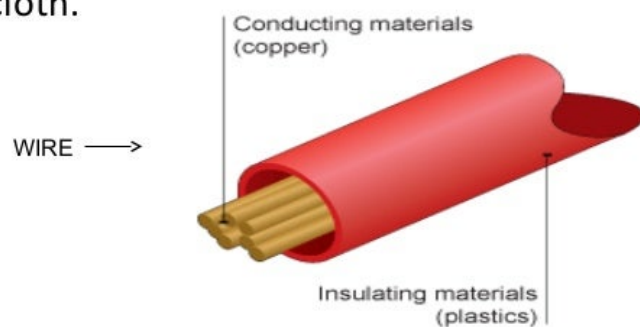


INSULATOR

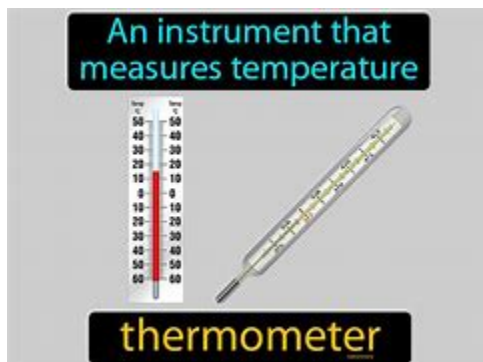
Are non -conducting material that do not easily allow current to pass through them

Insulators

- **Insulators: They don't easily allow** the flow of electric charges through them.
- Ex: glass, plastic, certain rubber materials, cloth.



THERMOMETER



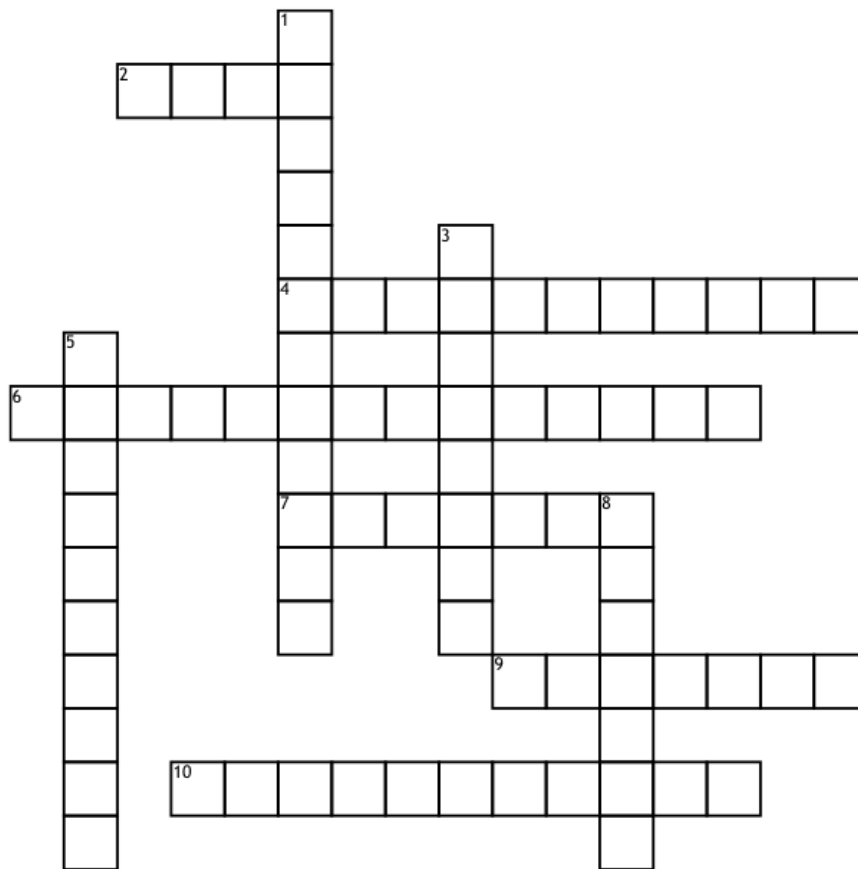
ACTIVITY: LEARNER HAS TO MATCH THE CONCEPT WITH THE DESCRIPTION

HEAT, TEMPERATURE, CELCIOUS SCALE, BOILING POINT, MELTING PONT, CONDUCTOR, INSULATOR, THEMOMETER, HEAT ENERGY

DESCRIPTION	CONCEPTS
1.An instrument that measure temperature	
2.A conductor is an object or type of material that allows the flow of An electrical current in one or more direction	
3.This is the temperature at which solid substance changes its state to become a liquid	
4.They don't allow the flow of electric charges through them	
5.Is the temperature at which a liquid change into a vapor	
6.Heat nergy is the form of energy transfer refers to energy contained within a system	
7.Is the transfer of energy from one body to another as a result of a difference in temperature	

Name: _____ Date: _____

Temperature Crossword



Across

- 2. 37 degrees celsius is our ... temperature
- 4. A tool used to measure temperature
- 6. The unit of measurement we use in SA
- 7. What we call the highest temperature
- 9. 100 degrees celsius is ... point

- 10. A measure of the warmth or coldness of an object or substance

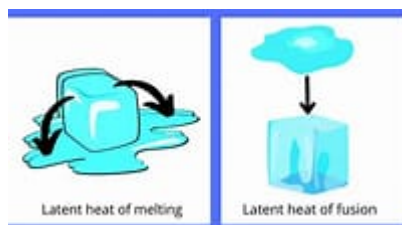
Down

- 1. If we experience very cold temperatures we can get ...
- 3. 0 degrees celsius is ... point
- 5. If we experience very hot temperatures we can become...
- 8. What we call the lowest temperature

PROCESS OF BOILING AND MELTING POINT

About 694 000 results

1.



Melting and boiling are processes in which these noncovalent attractive interactions are disrupted. The stronger the intermolecular attractive forces, the more energy that is required to break them apart. That energy is in the form of heat, e.g. a higher temperature must be reached for the melting or boiling to occur.

NB *FROM THE SMARTBOARD: DEMONSTRATION OF HOW TO READ A THERMOMETER MUST BE VIEWED

NB* DEMONSTRATION ON HOW HEATING AND COOLING CAN CHNGE MATERIALS

ACTIVITY: INVESTIGATION

MATERIAL AND APARATUS	METHOD OF THEIR OWN IS EXPLAINED BY THE LEARNER TO NVESTIGATE THE COOLING AND HEATING PROCESS
Glass beaker (2) Bunsen burner/one plate stove Tripod with gauze Tap water Thermometer Funnel Ice blocks/freezer	WHAT NEEDS TO HAPPEN WATER TO MAKE IT FREEZE? WHAT NEEDS TO HAPPEN TO MAKE IT BOIL ?

	(10)

2. State the 3 stages of matter and give examples

--	--	--

(6)

3.Can you indicate How do you measure the temperature for the above experiment

FOR THE HOT PROJECT	FOR THE COLD

(4)

4.What is the temperature for

HOTTER STATE _____	FREEZING STATE _____
---------------------------	-----------------------------

(2)

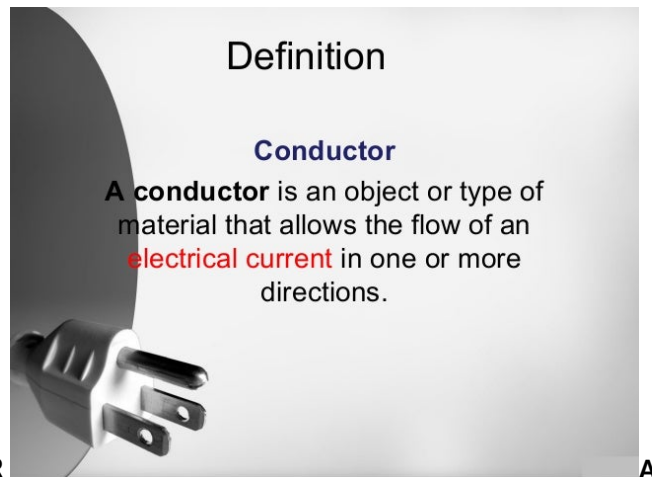
BEFORE YOU COMPLETE THIS ACTIVITY LET US OBSEVE YOUR INVESTIGATION ON HEATING AND COOLING PROJECT AND WRITE YOUR OBSERVATION IN THIS TABLE

ELAPSED TIME (MINUTES) ON 1.HEATTING 2.COOLING	TEMPERATURE OF WATER

1.	
2.	

(4)

INVESTIGATION ON CONDUCTORS AND INSULATORS (REMEMBER THIS PICTURE

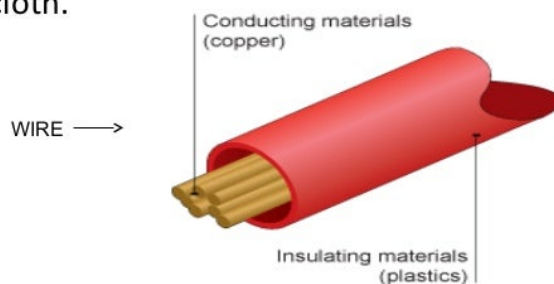


CONDUCTOR

A

Insulators

- **Insulators:** They don't easily allow the flow of electric charges through them.
- Ex: glass, plastic, certain rubber materials, cloth.



NB(GOOGLE)

ACTIVITY: INVESTIGATING CONDUCTORS AND INSULATORS

HOW WILL YOU KNOW THESE MATERIAL CONDUCT ELETRICITY?

CONDUCTORS AND INSULATORS



Make It Easy Education



CONDUCTORS (explain and give examples)	INSULATORS (explain and give examples)
1.	1.
2.	2.
3.	3.

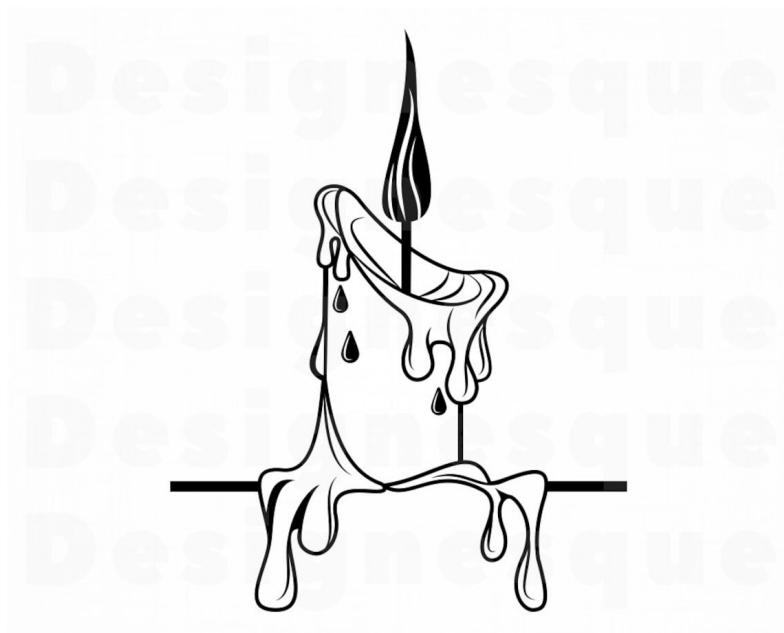
NB (OUR LEARNERS CAN NOT CONNECT ELECTRIC WIRES THEY CAN WATCH VIDEO ON SMART BOARD THEN ANSWER

VIDEO

VIEWED*****

1.What type of material did not light up ?

2.What does that mean?



3. Write a shot story what is happening with this candle

4. What was the mistake here

5. Do you think its wise to put a candle like this and what will happen when it reaches the final stage

[HOW TO USE A THERMOMETER](#)

Search instead for [HOW TO USE A THEMOMETER](#)

Search Results

Featured snippet from the web

Turn on the digital thermometer. Place the thermometer tip under your tongue. Close your mouth around the thermometer for the

recommended amount of time or until the thermometer beep indicates it's done. Remove the thermometer and read the number.



MEASURES TEMPERATURE

A thermometer is an instrument that measures temperature. It can measure the temperature of a solid such as food, a liquid such as water, or a gas such as air. The three most common units of measurement for temperature are Celsius, Fahrenheit, and kelvin.

WHY IS THERE ALCOHOL IN A THERMOMETER?

Unlike mercury, alcohol is less toxic and will evaporate away quickly if subjected to high temperatures. Therefore, it is used to measure low temperatures.

WHAT ALCOHOL IS USED IN A THERMOMETER?

ethanol

The liquid used can be pure ethanol, toluene, kerosene or isoamyl acetate, depending on manufacturer and working temperature range. Since these are transparent, the liquid is made more visible by the addition of a red or blue dye.

ANSWER THE FOLLOWING QUESTIONS

2.How to use a thermometer?

1.
2.
3.
4.

3. Why is there alcohol in a thermometer?

1.
2.

4. What alcohol is used in a thermometer?

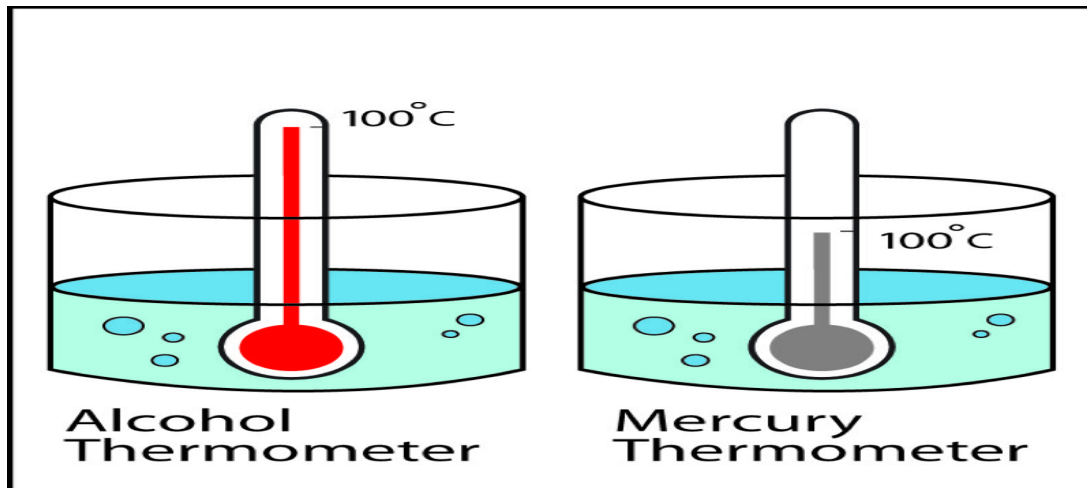
1.

5. What do you think will happen when the liquid level in a thermometer expands? (2)

6. What do you think will happen to the liquid level in the thermometer when it is put into a refrigerator?

HOW HEATING AND COOLING CAN CHANGE MATERIAL

Some materials expand on heating and some contract on cooling. Heating makes the particles (that form the material) expand or become loose. Cooling makes the particles (that form the material) contract or become tight. The amount of expansion differs in solids, liquids and gases.



DRAW YOUR OWN GRAPH AND PLOT THE FOLLOWING INFORMATION TO DEMONSTRATE YOUR UNDERSTANDING OF EXPANSION.

Material	Temperature increament
Brass	19mm
Iron	12mm
Steel	11mm
Platinum	10mm

EXAMPLE





1. Which material expands the most by looking at your graph?

2. Which material expands least/less?

TOPIC2: STORED ENERGY IN FUELS

WHAT ARE FUELS?



Any substance used to produce energy is called fuel. Any combustible substance which may be burnt to supply heat for practical applications without the formation of excessively objectionable byproducts.

Name all the fuels that can be burnt to create heat and light.

Matching concepts with description

Choose from petrol, wood, gas, paraffin, candle, electricity and fossil fuel

Description	Put the matching concepts
Comes from plants especially trees	
This fuel can be burnt to release stored energy	
Wax is burnt to provide light	
Is mainly used to fuel cars	
Is used to burnt in lamp and stoves	
Are non-renewable energy, were made from dead plants and animals and contain carbon.	
The flow of electrical power or	

To keep a multi fuel stove burning:

- **Build and light best fire possible to get the fire going well and provide a bed of hot coals.**
- **Warm up the floor if cold to get maximum draft on the stove**
- **Ensure that the wood has low moisture content.**
- **Use hardwood over softwood for longer burning fires between loads.**
- **Regulate the airflow using vents**
- **Keep stove topped up with fuel, without overloading the stove.**

ACTIVITY: INVESTIGATION

AIM: TO FIND OUT HOW A CANDLE WILL BURN FOR WHEN GIVEN DIFFERENT AMOUNT OF OXYGEN

Material	Method
Candle, different size of glass bottle (small, medium, large) flat bottom bowl, water	1. Light the candle 2. Drip some wax in the middle of the bowl then mount candle in the wax 3. Pour small amount of water in the bowl around candle 4. Light the candle 5. Place a small bottle over candle and record time as to how long it takes until the candle goes out



Size of the glass jar	Time taken for candle to go out

Small	
Medium	
Large	

Questions	Answer
1. Which jar did the candle burn longer?	
2. In which glass jar did the candle burn out the fastest?	
3. Why did the candle go out once you put the jar over the candle.	
4. A candle that is allowed to burn freely in air will eventually burn down and go out. Why does it stop burning?	

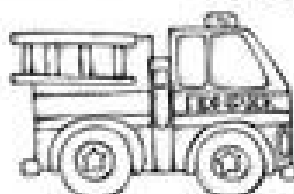
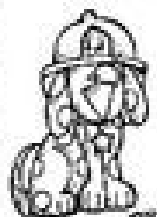
SAFETY WITH FIRE RULES

Basic Fire Safety Rules



Fire Safety Rules

1. Matches and lighters are not toys. Do not play with them. Give them to adults.
2. Do not hide during a fire. Let the firefighters see you and know where you are.
3. Stay low and crawl towards the nearest exit if you see and smell thick smoke.
4. Get out and stay out of burning buildings.
5. Stop, drop and roll if your clothes should catch on fire.
6. In case of an emergency, call 911 for help when you get to a safe place.
7. Keep safe and do not play with fire.



Fill in the word search puzzle below.



FIRE

WORD SEARCH PUZZLE

ALARM
BALL
BOAT
BRAND
DOOR
DRILL
ENGINE
ESCAPE
HOSE
HOUSE
IRON
LIGHT
PLACE
POWER
SALE
STATION
STORM
TOWER
TRAP
TRUCK
WARDEN
WATER
WOOD

E	S	U	O	H	D	N	A	R	B	A
L	S	M	R	O	T	S	N	O	R	I
B	A	L	L	T	N	T	H	O	S	E
O	L	L	R	E	H	A	F	D	R	E
A	E	U	D	G	L	T	R	E	N	R
T	C	R	I	A	R	I	W	I	E	E
K	A	L	R	A	L	O	G	T	I	W
W	L	M	P	L	P	N	A	R	E	O
E	P	A	C	S	E	W	O	O	D	T

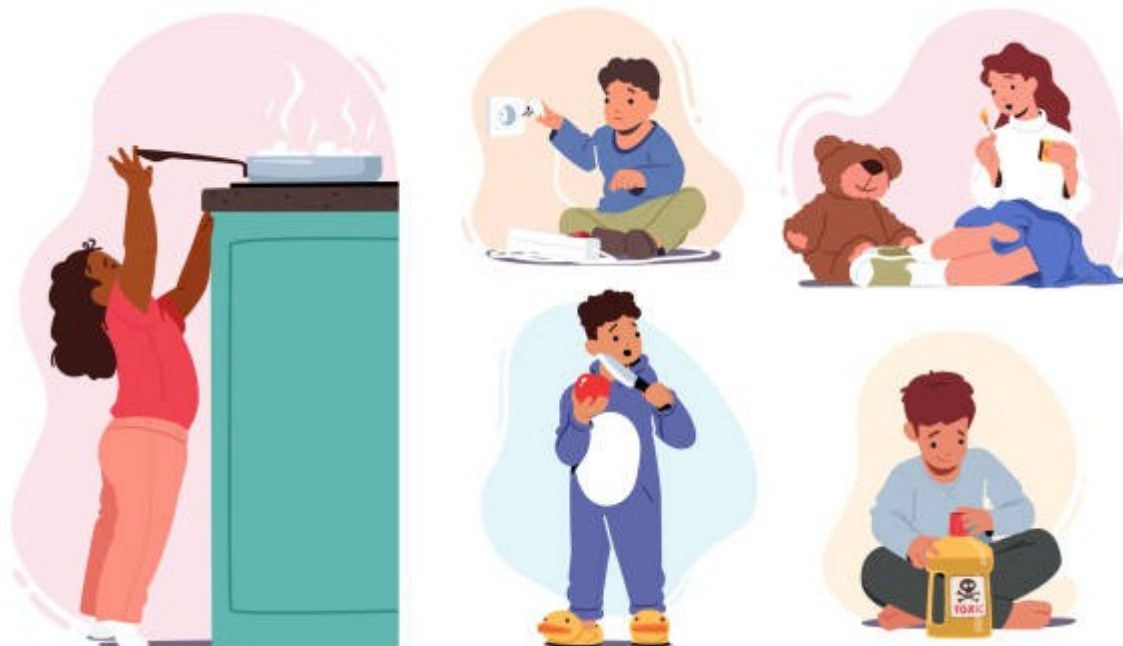
The words appear UP, DOWN, BACKWARDS, and
DIAGONALLY. Find and circle each word.



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Activity: DANGEROUS SITUATION INVOLVING FIRE INSTRUCTIONS

1. Below you have 4 different scenes



Name the danger and write the explanation of the danger.

Situation (danger)	Why is it dangerous?
1.	
2.	
3.	
4.	

Activity: what are you going to do in a case of fire (act out in groups)

Questions teachers asks the group to act out:

- **How to escape from burning building**
- **Not to open a door in a building that is burning**
- **What to do if clothes are on fire**
- **What to do if your friend is stuck in a burning building (15)**

5 members of the group
1.
2.
3.
4.
5.

Activity: CREATE A FIRE POSTER (20)

Group of 5

Materials needed:

- **Cardboard/ A4 page**
- **Colored pens/pencils**
- **Old magazines**
- **Scissors**
- **Glue**

Instructions

- 1. Rules on what to do incase fire start at school**
- 2. Steps to follow with pictures from old magazine**
- 3. How will you make sure no one is left inside the building**

Rubric

Description	1	2	3	4	5
Labels clear					
Pictures pasted with understanding					
Rules for fire management explained and show understanding					
Neatness					

Topic 4: ENERGY AND ELECTRICITY

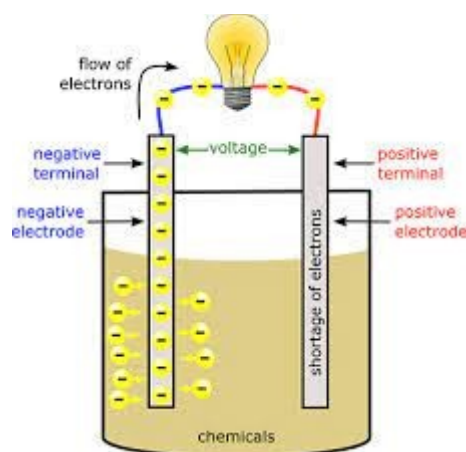
4.1 CELLS AND BATTERIES

KEYWORDS IN THE FOLLOWING TOPIC

- BATTERY**
- ELECTRIC CIRCUITS**
- ELECTRICAL COMPONENTS**
- STORED ENERGY**
- SOURCE OF ENERGY**

WHAT IS A BATTERY?

A battery is a device that stores energy and makes it available in an electrical form. A battery converts chemical energy into electrical energy. Or it is a device that produces electrical energy from chemical energy.

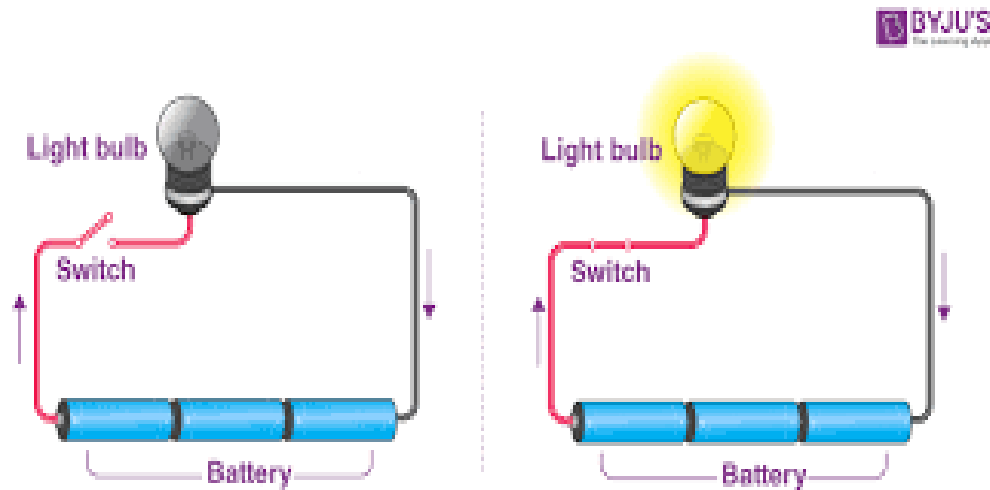


Electric circuits

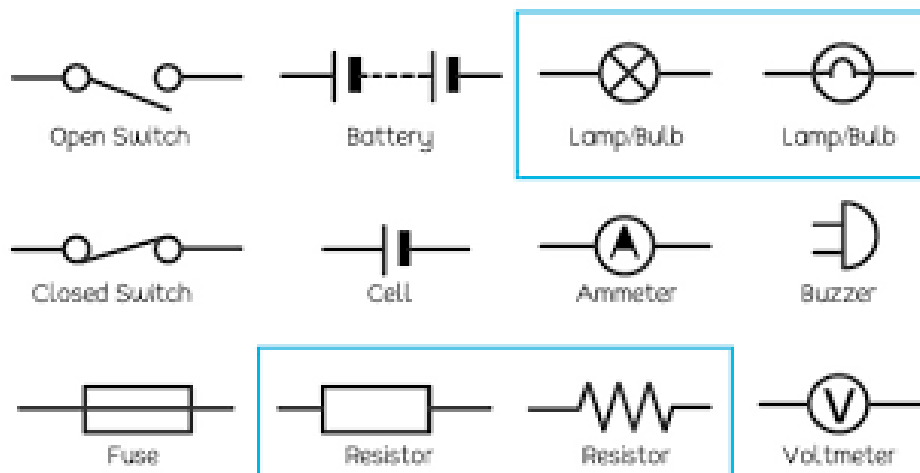
An electric circuit is a connection of components,

- voltage/current sources**
- resistors**
- inductors**
- Capacitors**

Where some power is supplied and then dissipated. One of the most important ideas of an electric circuit is that there is a source of power and a dissipater of power.



Symbols of the components



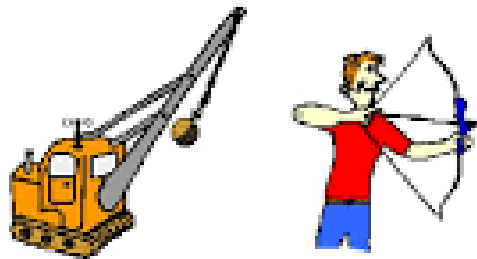
Four major components of an electrical system

- **Sources**
- **Conductors**
- **Loads**
- **switches**

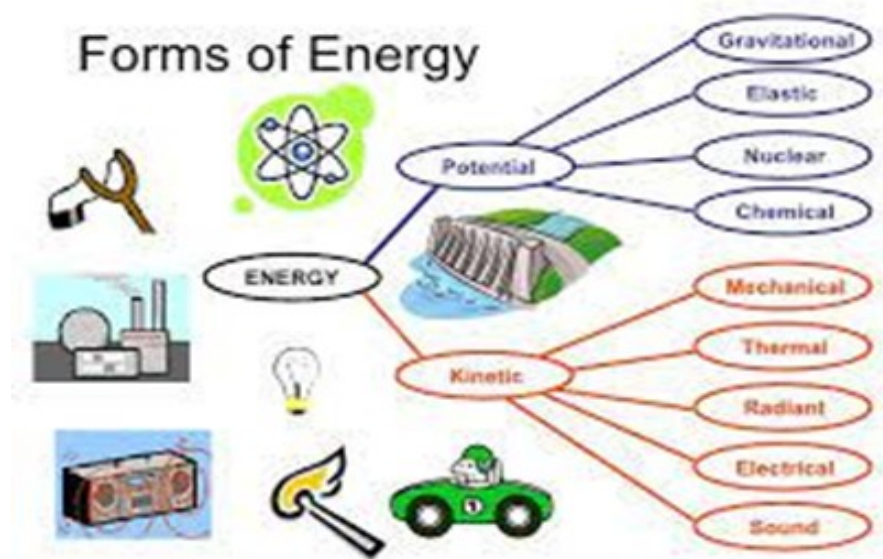


Potential energy

Potential energy is the stored energy of an object.



The massive ball of a demolition machine and the stretched bow possesses stored energy of position - potential energy.



Sources of Energy

Renewable

Energy sources that are easily replaced or available on a perpetual basis, if managed properly

Nonrenewable

Energy sources found in limited amounts. They are used up much faster than they can be replaced



Activity: match the concept with the description

- BATTERY**
- ELECTRIC CIRCUITS**
- ELECTRICAL COMPONENTS**
- STORED ENERGY**
- SOURCE OF ENERGY**

Description	Matching concept
1. it is a connection of components (voltage/ current sources, resistors, inductors and captors)	
2. a battery is the device that stores energy and make it available in an electrical form	
3. those are sources, conductors, loads and switches.	
4. That is the energy stored energy that has not been used	
5. Those falls on renewable energy sources that are easily replaced and non-renewable energy sources found in limited amounts.	

4. Different types of batteries and their uses

DIFFERENT TYPES OF BATTERIES

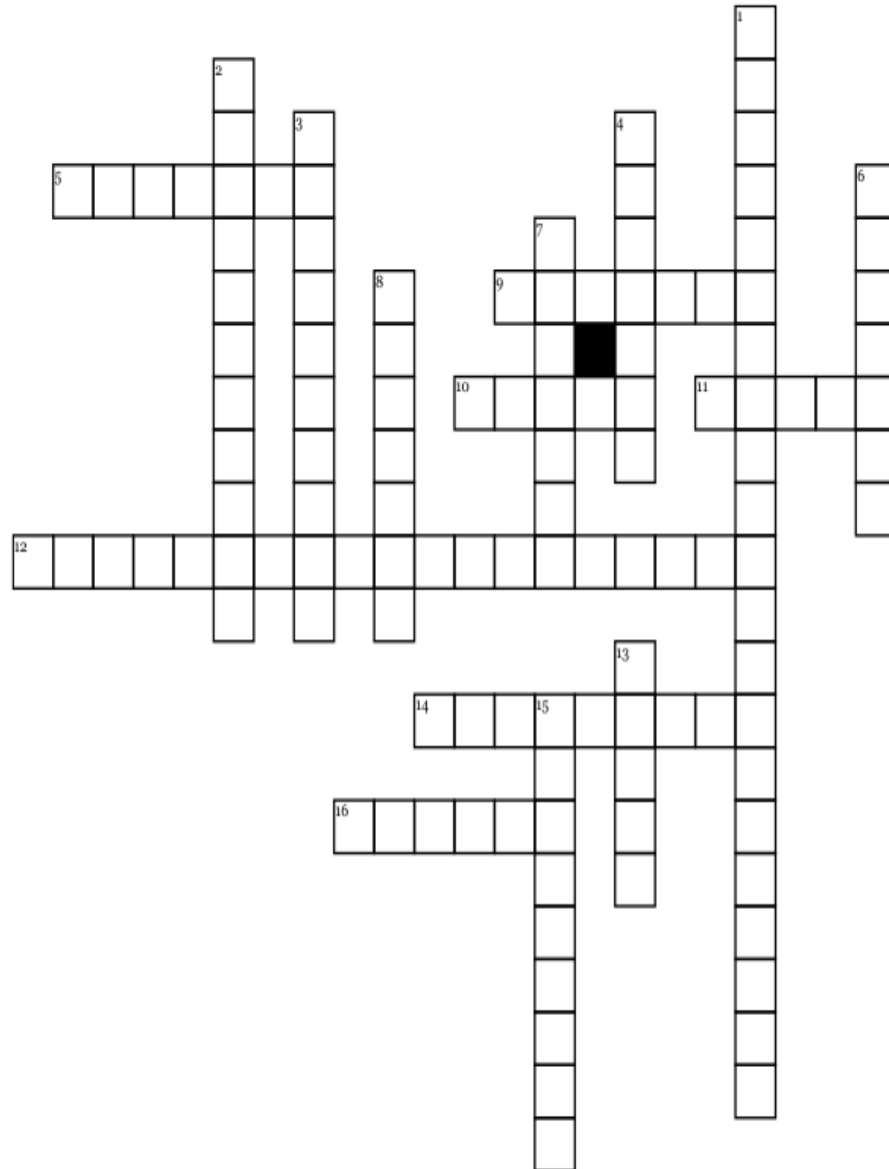


Battery applications

Battery Type	Key Attributes	Application & Battery Selection
Alkaline Cylindrical 	• Inexpensive • Widely Available • Moderate Drain Rate • AA, AAA, AAAA	Wide Variety of Portable Devices 
Lithium Cylindrical 	• High Performance • Shelf Life of 15 years • High Drain Rate • Extreme Temperature • High Reliability • Lightweight • AA, AAA	Higher Drain Rates/Long Service Life 
Lithium Coin 	• Small • Lightweight • Low Drain Rate • 7-10 Year Shelf Life • 1/3 sizes	Small, Low Energy Devices 

Name: _____

Battery Crossword



Across

- 5.** Type of battery used in a car
9. Measure of potential difference
10. Electrical -> Chemical energy
11. Positively charged electrode
12. Difference between cathode and anode

14. Mechanical -> Electrical Energy

16. Measure of Current

Down

- 1.** Chemical Cell
2. A liquid or gel that allows flow between the 2 electrodes
3. A battery converts chemical energy into _____ energy

4. An object that stores energy

6. Type of battery in a flashlight

7. Strength of Battery

8. Negatively charged electrode

13. Electrolyte in a dry cell

15. A cathode or anode

